

a

Stage game (one round)

		<i>opponent</i>	
		cooperate	defect
<i>focal player</i>	cooperate	2, 2 CC	10, 0 CD
	defect	0, 10 DC	6, 6 DD

$T > R > P > S$ and $2R > T + S$ greed = $(T - R)/(T - S) = 0.20$ fear = $(P - S)/(T - S) = 0.40$

payoffs shown at $\lambda = 1$; every cell is multiplied by $\lambda \in \{0.1, 1, 10\}$

Horizon condition differs by arm: the Gemini games were run with the round count disclosed to the agents, the other three arms with the horizon hidden.

b

Fully crossed design

model	Claude	Gemini	GPT-4o	Mistral	×4	
horizon	unknown	known	unknown	unknown	confound	
payoff scale λ	×0.1	×1	×10		×3	
language	EN	FR	VI	ZH	AR	×5
persona pairing	C-C	C-S	S-C	S-S		×4
replicate						×10

2,400 dyads × 10 rounds × 2 agents = 48,000 binary decisions